

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary and Renal

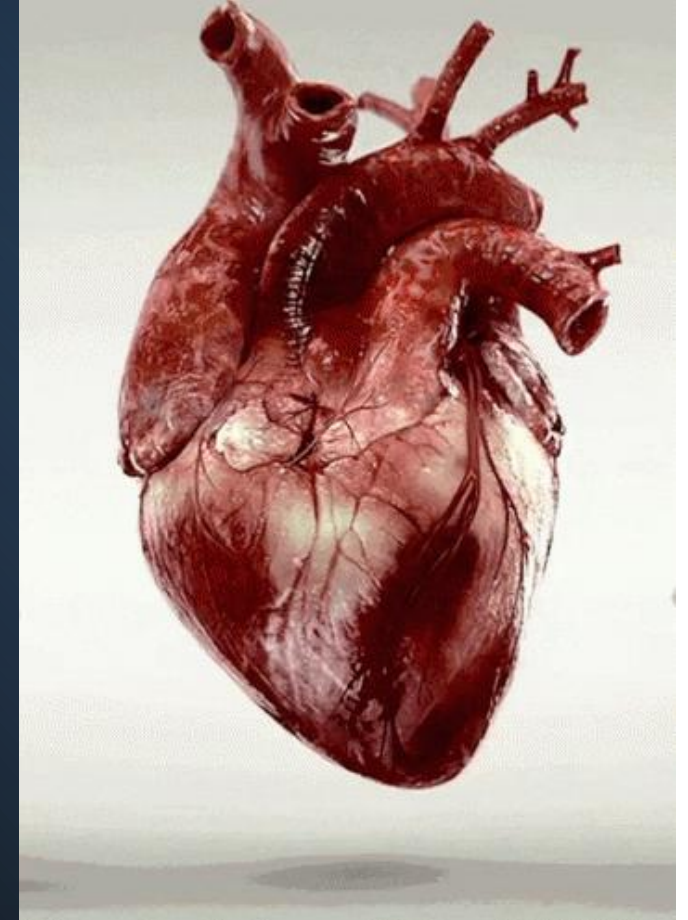
Lecture: CPR 07 Mechanics of the Cardiac Pump

Dr. Gabrielle Walcott-Bedeau

Associate Professor

Department of Physiology, Neuroscience and
Behavioral Sciences

Email: gwalcott@sgu.edu



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Recommended readings

Chapter 13: Mechanics of the Cardiac Pump. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

Cardiac cycle as a wigger's diagram and pressure volume loops must be reviewed prior to this lecture.

Remember to start your practice questions: BRS review, Guyton and Hall review, Pretest, Sakai practice questions and Clinical Key

Slides with blue background are the supplemental information associated with the previous slides and thus, might not be explained during the lecture time. However, please read the slides for clarification as they are equally important to know.

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0096 Draw and describe the length-tension relationship in a single cardiac cell. Correlate length, tension, and velocity of shortening with the intact ventricle characteristics of end diastolic volume and pressure.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0097 Define preload and explain why ventricular end-diastolic pressure, atrial pressure and venous pressure all provide estimates of ventricular preload. Explain why ventricular end-diastolic pressure provides the most reliable estimate.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0098 Define afterload and explain how arterial pressure increases or decreases afterload.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0099 Define contractility.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0358 Explain how the calcium transient differs between cardiac and skeletal muscle and how this influences contractility.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0100 Describe the difference between cardiac performance and cardiac contractility and the impact of changes in preload, afterload, and contractility in determining cardiac performance.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0101 Explain how changes in sympathetic activity alter ventricular work, cardiac metabolism, oxygen consumption and cardiac output.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0102 Use the law of Laplace to predict the effects of pressure, radius and wall thickness on ventricular function.

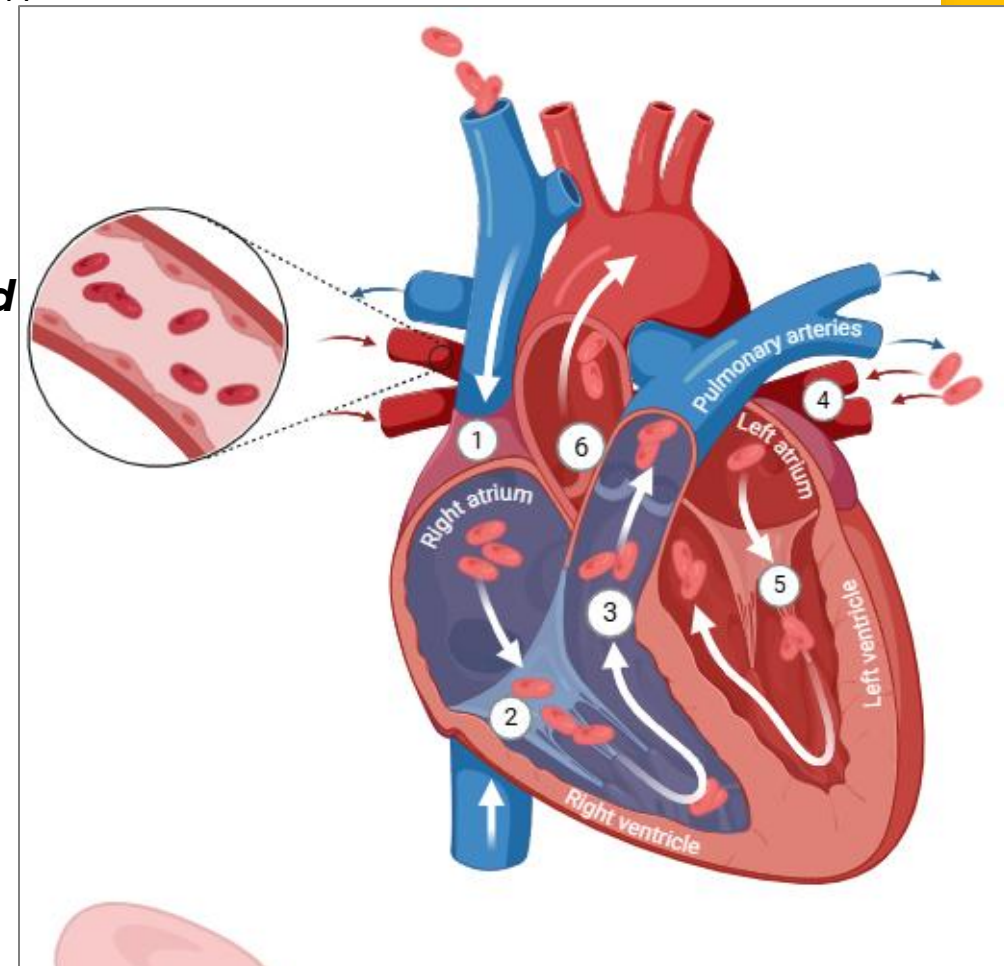
Abbreviations

- CO – cardiac output ($CO = SV \times HR$)
- Ejection fraction ($EF = SV/EDV$)
- EDV – end diastolic volume
- EDPVR – end diastolic pressure volume relationship
- ESPVR – end systolic pressure volume relationship
- LVEDV – left ventricular end diastolic volume
- RAP – right atrial pressure
- SV – stroke volume ($SV = EDV - ESV$)
- VR – venous return
- SVR/TPR –systemic vascular resistance/total peripheral resistance
- MAP – mean arterial pressure
- PP – pulse pressure
- SBP – systolic blood pressure
- DBP – diastolic blood pressure



Points to consider – the big picture

- Myocardial contraction is caused by electrical activity of heart. (*Refer to lectures on cardiac action potential and ECG*)
- The mechanical 'pumping' performance is determined by the number of cross-bridge cycles during contraction. (*Refer to MSK lectures on excitation-contraction coupling*)
- During diastole, the heart fills with blood (**preload**).
- During systole, the heart (**contracts**) pressurizes and ejects blood.
- Ejection occurs when the pressure/load surpasses the arterial pressure/load (**afterload**) allowing the outlet valve to open.
- Heart **contracts** by **developing force (isometric contraction) and shortening (isotonic contraction)**
- **Determinants of myocardial performance:**
 - Preload
 - Afterload
 - Contractility
- Students must understand:
 - Length-tension relationship
 - Force-velocity relationship
 - Frank-Starling mechanism
- Clinical application - Myocardial performance changes:
 - e.g. exercise – increases pumping activity to meet increased demands
 - e.g. ischemia/heart failure – damage impairs pump activity



Lecture Outline

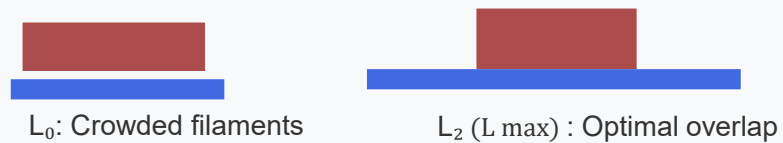
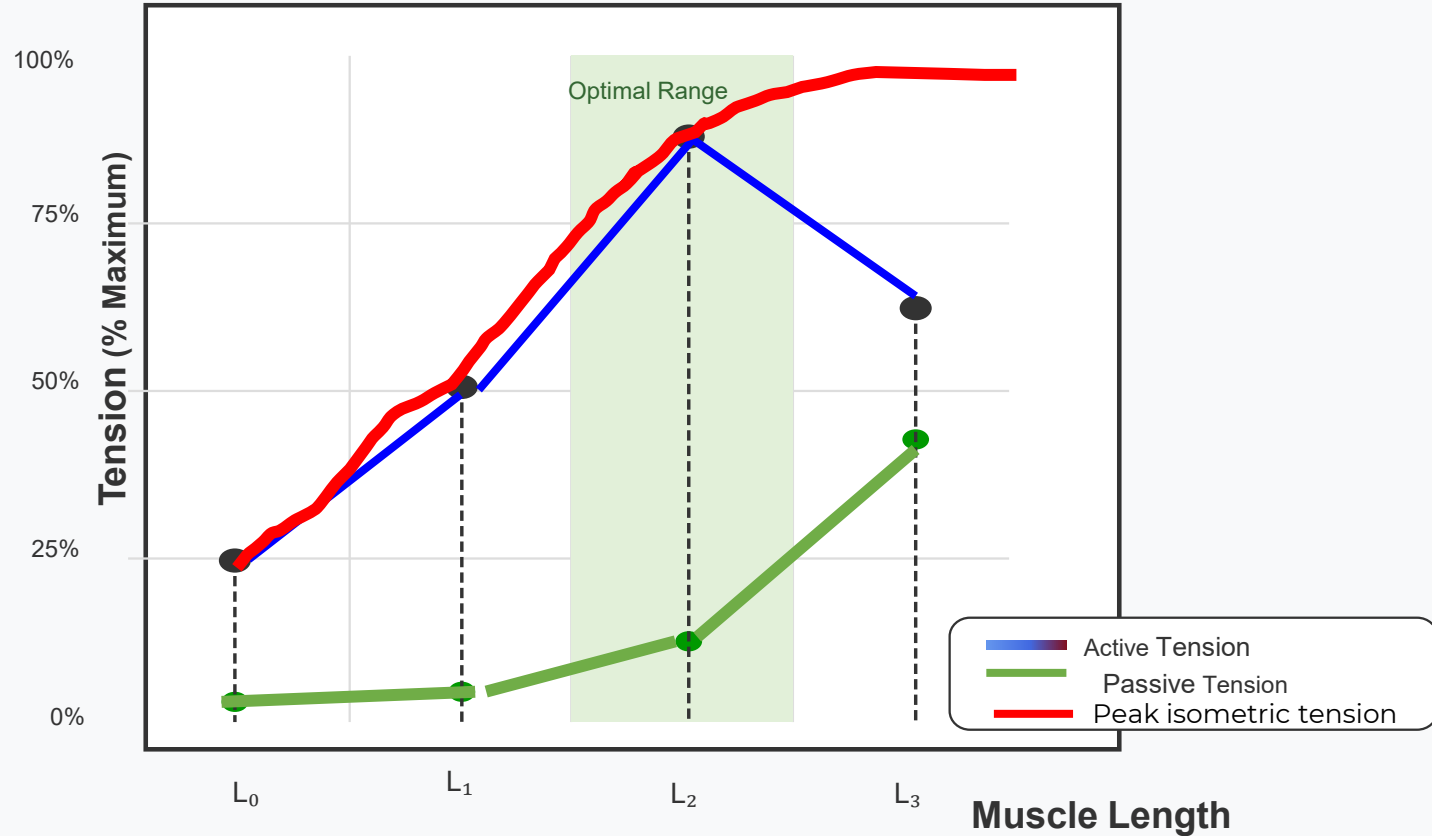
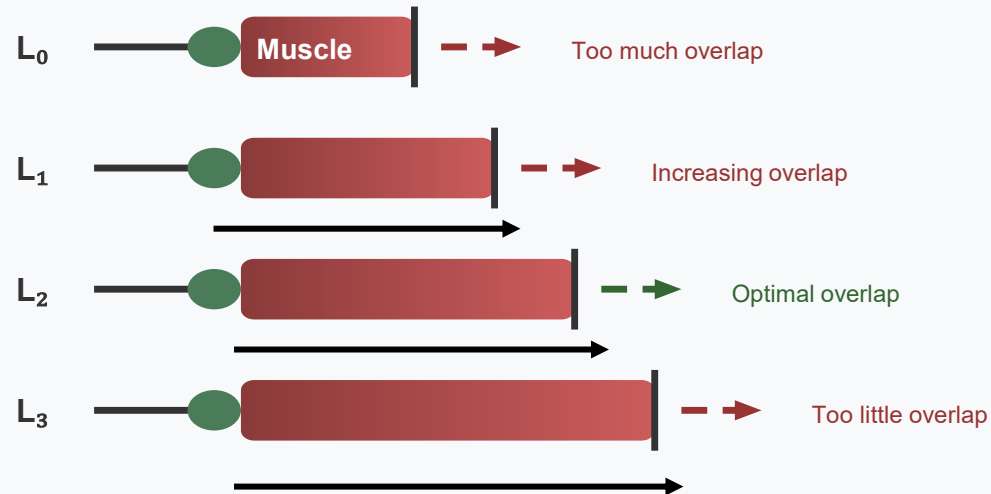
- **Length-tension relationship – cardiac muscle**
- **Force-Velocity relationship – cardiac muscle**
- **Main concepts – discuss ventricular function, with changes in preload, afterload and contractility changes**
- **Clinical correlations**

Mechanics of cardiac muscle contraction

Length-tension relationship in cardiac muscle (Preload)

In 1895 – Otto Frank first described the cardiac muscle length-tension relationship

Muscle States



Sarcomere Structure at Different Lengths

Myosin filaments (red) and Actin filaments (blue)

Note: Passive tension (green) increases as muscle is stretched, becoming significant at longer lengths.

Explanation of previous slide

Experimental setup:

- Muscle fixed at one end, attached to a force transducer at the other.
- Length (L_0 – L_3) adjusted to vary preload (stretch). Green line
- Stimulation triggers contraction; force measured as tension. Blue line

Definitions:

- L_0 – Unstressed length (no passive tension)
- Passive Tension - Force required to stretch a resting muscle to different lengths.
- Active Tension - When a muscle is stimulated to contract whilst its length is held constant, it develops an additional force called *active tension*.
- Total tension (Peak isometric tension) = Passive + active tension Red line

The amount of active tension developed by a cardiac muscle during an isometric contraction depends on the initial myocardial muscle length. Stretching the resting myocardial muscle fibre before it contracts will increase the amount of tension it can develop – up to a point.

$L_{\max}$ is the optimum muscle length at which maximum tension can occur. Cardiac muscles operate at where the level of force of contraction is 50% $\max$ tension.

- Small changes in myocardial muscle fibre length $\rightarrow$ large changes in active tension (red line)
- Cardiac muscle is much stiffer than skeletal m (resting tension is much greater)

Summary 1 – Effects of Preload

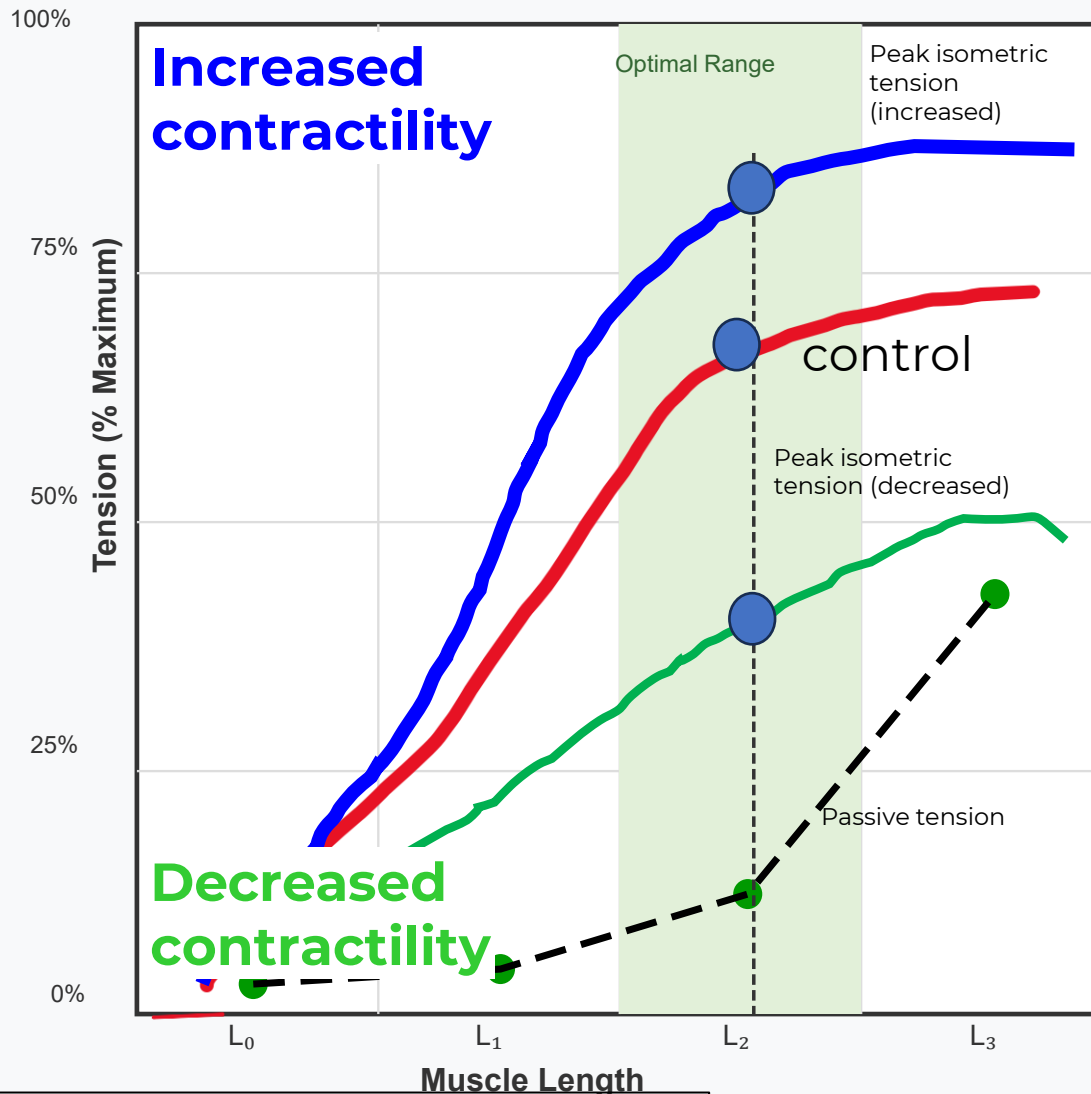
force of contraction is directly proportional to the initial length of the muscle fiber
the greater the stretch of the ventricular muscle, the more powerful the contraction

1. If resting muscle fibre is stretched, tension developed increases → ↑ force of contraction
2. As muscle length is ↑ ed – length-dependent change in sensitivity of the myofilaments to calcium ↑ - ability of TN-C to bind Ca is ↑ ed
3. Results in optimal overlap between actin and myosin filaments – ↑ muscle contraction
4. Resting muscle fibre length is increased by increasing the load on the muscle (before it contracts). This load – termed **“PRELOAD”**
5. ↑Preload → ↑ Force of Contraction

How else can the force of contraction of cardiac muscle be increased?
(i.e. apart from increasing preload)

Mechanics of cardiac muscle contraction

Length-tension relationship in cardiac muscle (Contractility)



From a given length L_2 , resting/passive tension remains the same, i.e. preload is unchanged

Peak isometric tension changes because active tension changes

- Blue line – force increases
- Green line – force decreases
- A change in contractility means that the muscle will, at every length, develop more/less isometric tension when stimulated than it would if agent was not present.

This suggests change in active force is change in ability of muscle to contract once stimulated.

Changes in contractility involve a **shift** from one peak isometric tension curve to another.

- Blue line- improved contractility
- Green line –reduced contractility

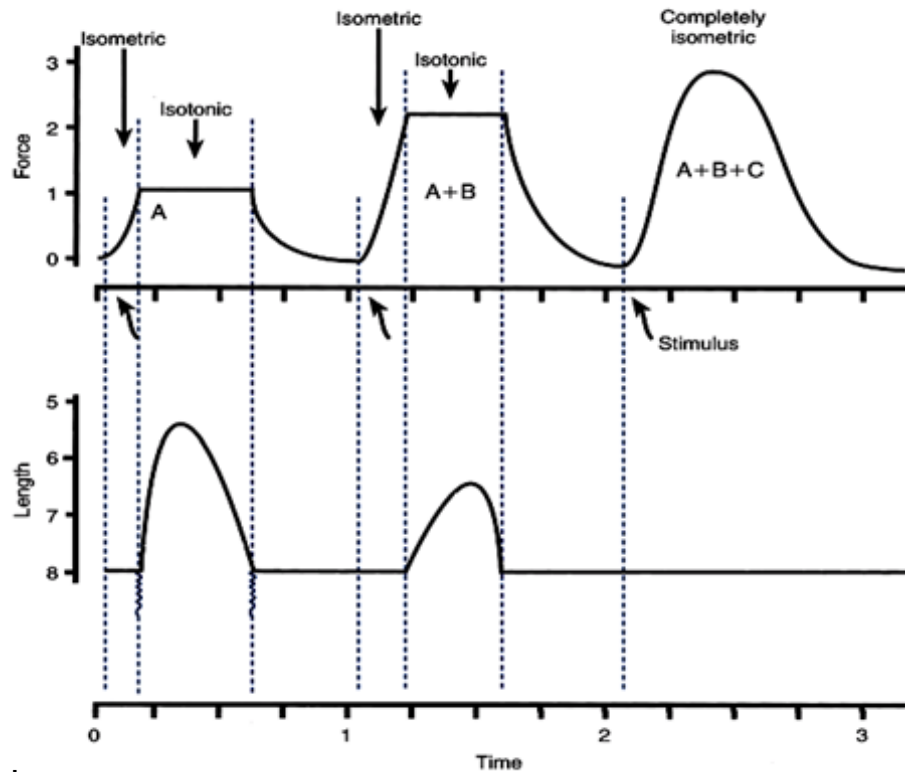
Inotropism (contractility) - a change in active force development in the **absence** of a change in preload (i.e. resting muscle length)

- **Positive inotropic factors** - increase contractility – Norepinephrine
- **Negative inotropic factors** - decrease contractility – Calcium channel blockers

Mechanics of cardiac muscle contraction

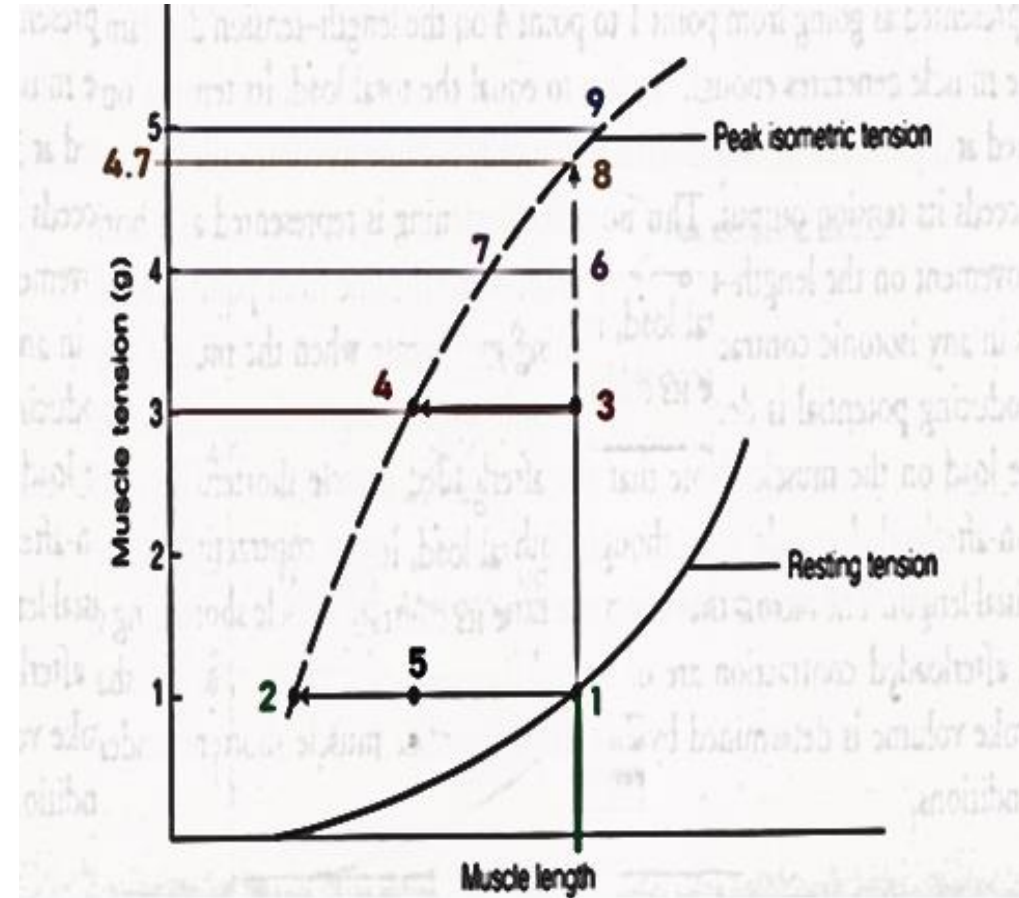
Length-tension relationship in cardiac muscle (Afterload)

Another way of looking at it.



Graph 1

- Muscle fibers contract against an “afterload”,
- Increased afterload – increased isometric tension
 - Increased afterload – decreased shortening



Graph 2

Mechanics of cardiac muscle contraction

Length-tension relationship in cardiac muscle (Afterload)

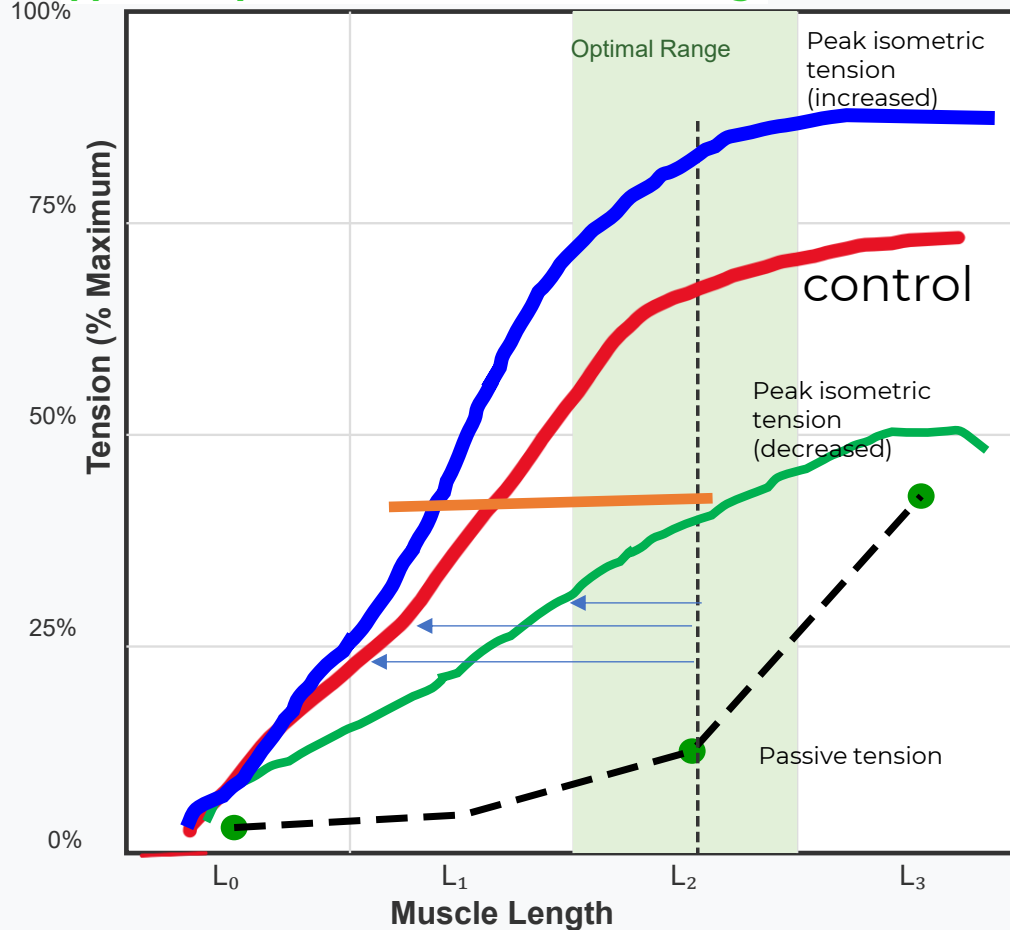
Explanation of previous slide graph 2.

- A muscle at length 1, can generate a peak isometric tension (contractile potential) up to point 8.
- A muscle at length 1, can generate an **isotonic** tension up to point 2, notice the shortening of length with contraction
- If that muscle at length 1 - has to contract against an afterload of 2 g, then the muscle can generate an isometric tension to point 3.
- At point 3 - the muscle would have developed adequate tension (3g) to overcome the afterload of 2g and the preload of 1g (already on the muscle).
- At point 3, the muscle would start to shorten (isotonically contract). The amount of shortening is limited by the peak isometric tension curve (point 4). As the muscle is getting shorter and shorter the amount of tension it is capable of developing is reduced.
- **The greater the afterload, the more isometric tension the myocardium has to develop before it can shorten;**
- **The greater the isometric tension developed then the amount of shortening that can occur is reduced.**
- Therefore, if it has to develop a tension of 3 g it shortens from 3 to 4, if it has to develop a tension of 4 g it shortens from 6 to 7. Compare the difference in lengths 6 to 7 is less than 3 to 4.
- At point 8, the peak tension is 4.7g.
- If the muscle has to contract against 5g without a change in preload/resting length, then this is not possible.
- Peak isometric tension curve sets the limit for shortening

The effect of Inotropic factors on afterload

(+) Inotropic effect – more shortening

(-) Inotropic effect – less shortening



The change in length indicates muscle shortening or isotonic contraction.

Positive Inotropic effect

- Blue line
- large changes in length, greater shortening,
- sympathetic activity via NE and β activation
- peak isometric tension curve shifts up and to the left from control

Negative Inotropic effect

- Green line
- small changes in length, less shortening
- Calcium channel blockers, MI,
- Peak isometric tension curve shifts downwards and to right from control

The extent of shortening of muscle changes at any given set of preload and afterload. (note horizontal arrow)

Summary 2 – effects of contractility and afterload

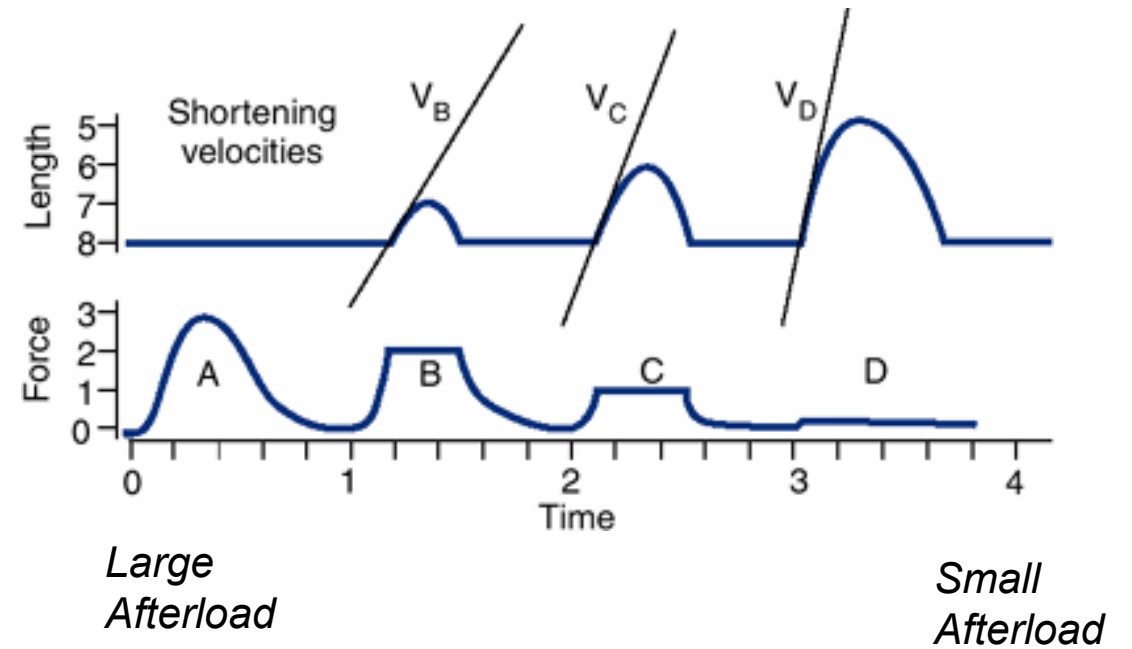
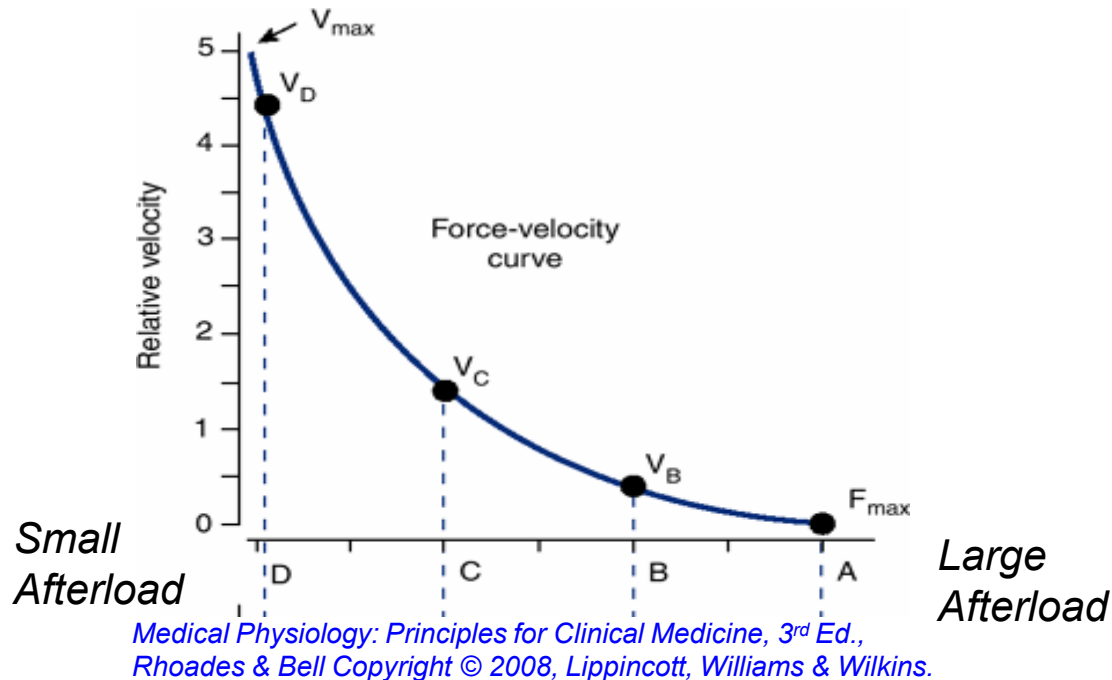
1. Isometric tension/contraction – maximum force/tension at given length/preload
2. Isotonic tension/contraction – tension remains the same, whilst muscle length changes shortening to overcome a given force/afterload. the myocardial fibres must first produce an isometric tension equal to the afterload, once this has been achieved, the myocardium will shorten.
3. Length-tension curve changes:
 1. Isometric and isotonic contraction increases with sympathetic activity via NE and β_1 activation. Positive inotropes.
 2. Isometric and isotonic contraction decreases with reduced availability of calcium e.g. channel blockers. Negative inotropes
 - +ve inotropes - improve the rate of crossbridge cycling.
 - +ve inotropes - $\uparrow$ the amount of shortening possible
 - +ve inotropes - $\uparrow$ myocardial contractility,
 - +ve inotrope - shift the peak isometric curve up and to the left.

Lecture Outline

- Length-tension relationship – cardiac muscle
- Force-Velocity relationship – cardiac muscle
- Main concepts – discuss ventricular function, with changes in preload, afterload and contractility changes
- Clinical correlations

Force-Velocity relationships

Comparing isotonic contractions for varying afterloads, if preload is kept constant

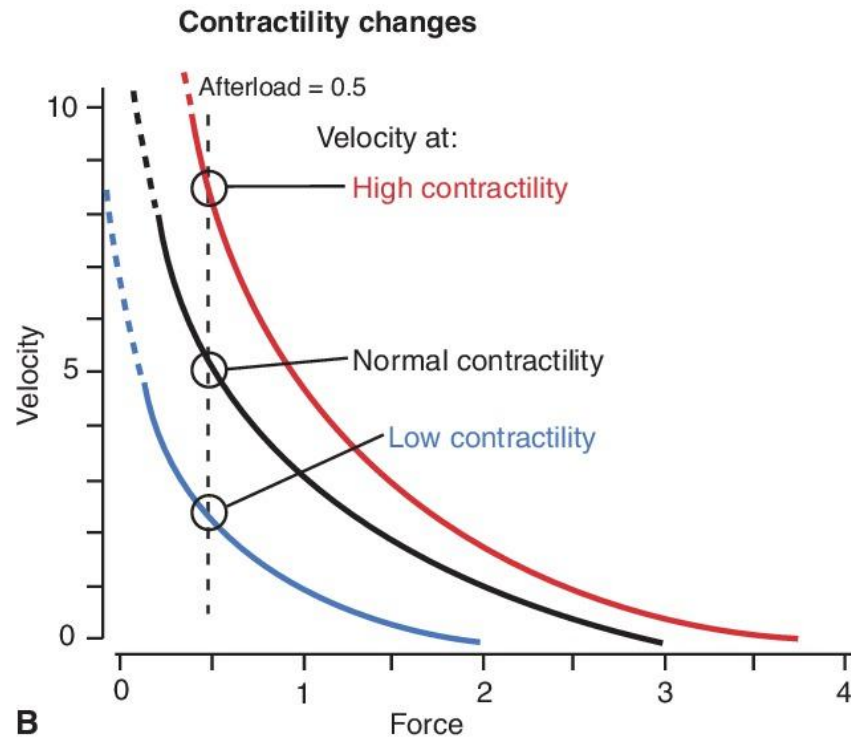
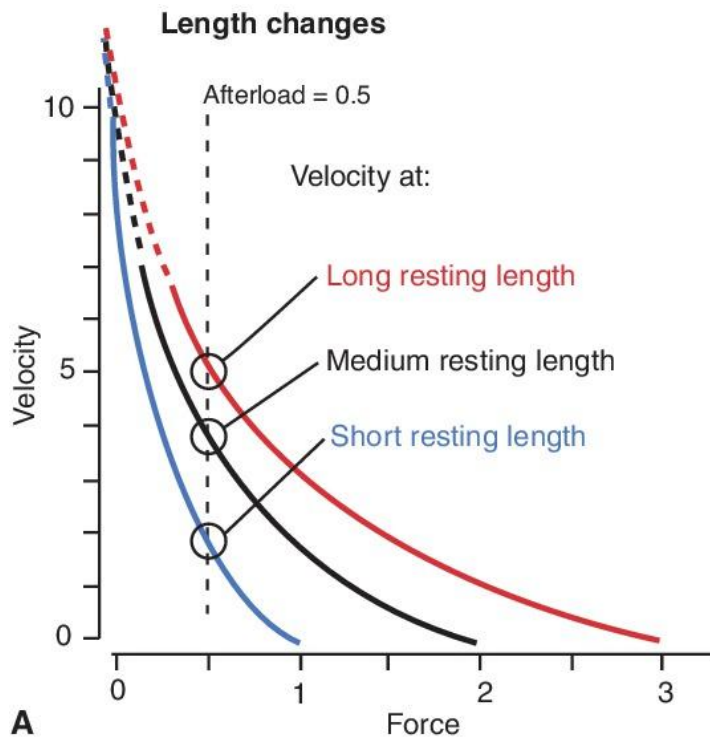


When afterload is large (A) no shortening occurs.

The smaller the afterload the faster the shortening velocity

Force-Velocity relationships

Comparing velocities for varying preload and contractility, if afterload is kept constant



(A) Decreased starting length (with constant contractility) produces lower velocities of shortening at a given afterload but does not affect the theoretical maximum velocity of shortening ($V_{\max}$). $V_{\max}$ is always an extrapolated estimate because it is impossible to make a direct measure of a zero-force contraction at each length.

(B) Increased contractility produces increased velocity of shortening at a constant muscle length; $V_{\max}$ is proportional to the inotropic state.

Summary 3 – Effects of velocity

- The velocity of muscle shortening depends on the load acting on the muscle
- The greater the afterload, the more isometric tension must be developed, and the amount of shortening will be smaller. The amount of shortening is limited by the peak isometric curve.
- The shortening velocity is the speed at which a muscle changes length during a contraction.
- The greater the afterload, the smaller the velocity of shortening.
- The greater the preload (the resting length of the muscle) the greater the velocity of shortening.
- The greater the contractility, the greater the velocity,
- +ve inotropes, $\uparrow$ contractility will $\uparrow$ $V_{\max}$.

Link to big picture:

During diastole, the ventricular fibers are stretched to develop a degree of tension to begin contraction. To eject blood, the heart has to develop a tension (isometric contraction) that will cause the ventricular pressure to exceed the aortic pressure so that the valves will open, and then the myocardial fibres will have to shorten (isotonic contraction) to eject the blood.

The major influential factors affecting stroke volume are:

- PRELOAD - Changes in myocardial muscle fiber length
- AFTERLOAD - the forces the muscle must overcome to open the valves and to eject a given volume of blood
- INOTROPIC STATE – contractile ability of the heart

High Yield definitions

preload, afterload and contractility *

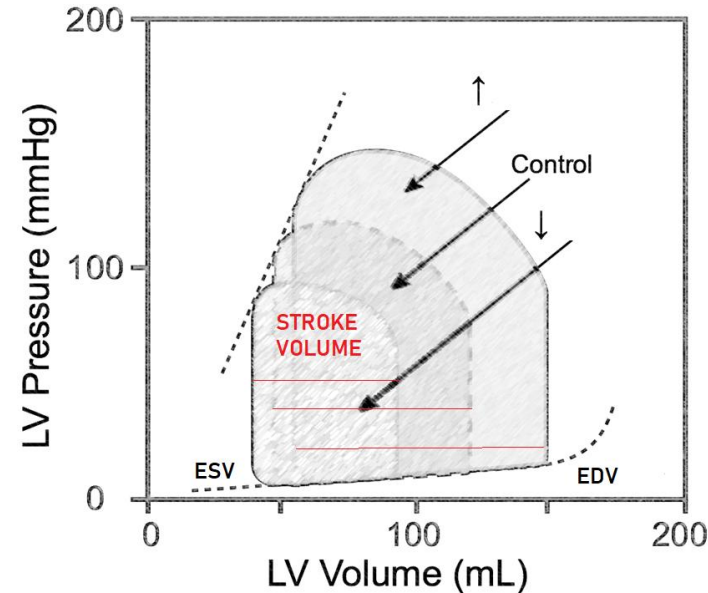
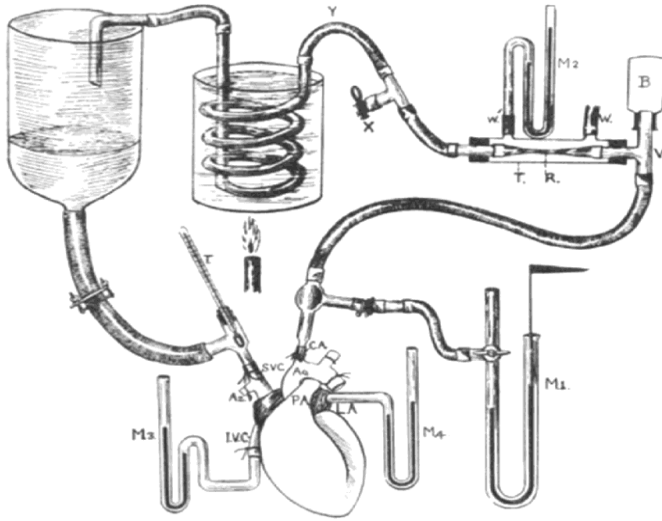
- **Preload:** is the degree of tension or load on the ventricular muscle when it begins to contract/at the end of diastole. It cannot be measured directly. Instead, indices such as left ventricular end-diastolic volume and left ventricular end-diastolic pressure are measured directly. The preload can be explained on the basis of change in sarcomere length.
- **Students must understand length-tension relationship and Frank Starling Mechanism to appreciate this function.**
- **Afterload:** the load that the heart must eject blood against. It is the pressure in the aorta leading from the ventricle. The afterload of the ventricle corresponds to the systolic pressure described by the phase III curve of the volume-pressure diagram. Most referred 'marker' of afterload is systemic vascular resistance also called total peripheral resistance. Although mean systolic ventricular pressures
- **Contractility:** change in performance at a given preload or afterload. Change in force of contraction at any given sarcomere length. Acute changes in intracellular calcium and drugs can alter the contractility. Contractility affects the rate at which a muscle can develop active tension. Ejection fraction is the best index of contractility.
- Students must understand ventricular function curves to appreciate the effects of preload versus contractility and afterload.

Lecture Outline

- **Length-tension relationship – cardiac muscle**
- **Force-Velocity relationship – cardiac muscle**
- **Main concepts – discuss ventricular function, with changes in preload, afterload and contractility changes**
- **Clinical correlations**

Ventricular function – Preload – Frank Starling's mechanism ★

In 1914 – Ernest Starling expanded the concept by describing the pressure-volume relationship



Law of the Heart (Frank-Starling Law):

The stroke volume of the heart increases in response to an increase in end-diastolic volume (preload), when all other factors remain constant. This occurs because increased ventricular filling stretches the cardiac muscle to a more optimal length on the length-tension curve.

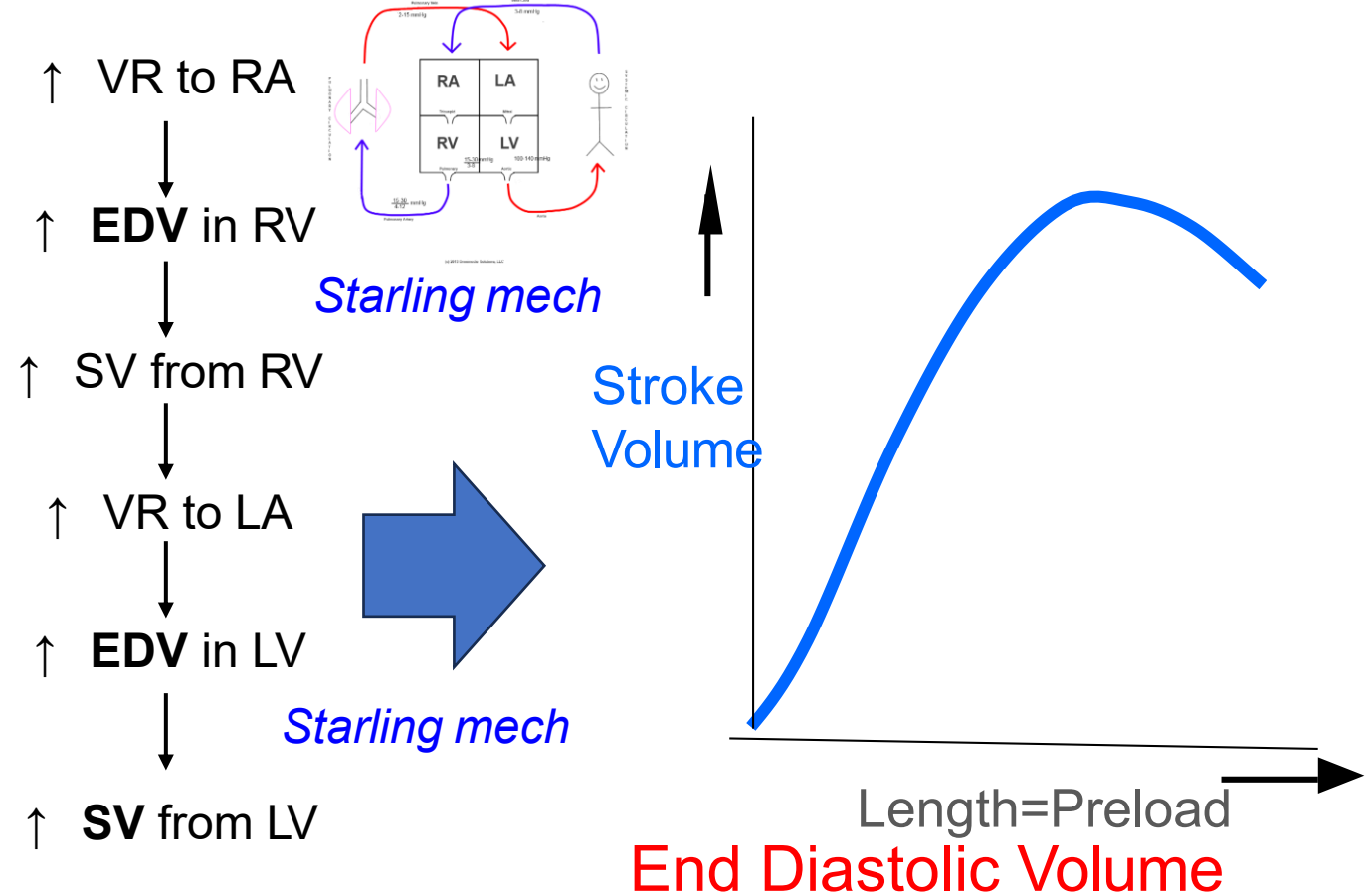
Frank Starling's Law/mechanism of the Heart: The intrinsic ability of the heart to adapt to increasing volumes of inflowing blood.

- The stretch/increased length allows **↑ Ca sensitivity of TN-C**, and **↑ actin-myosin interaction**.
- The energy of contraction of a cardiac muscle fiber is proportional to the initial muscle fibre length.
- Changes in myocardial fibre **length** are measured as changes in **blood volume**
- Changes in **tension** are measure as changes in **pressure** in myocardium.
- Stroke volume = EDV – ESV
- Stroke work = intraventricular pressure needed to eject stroke volume

Ventricular function – Preload

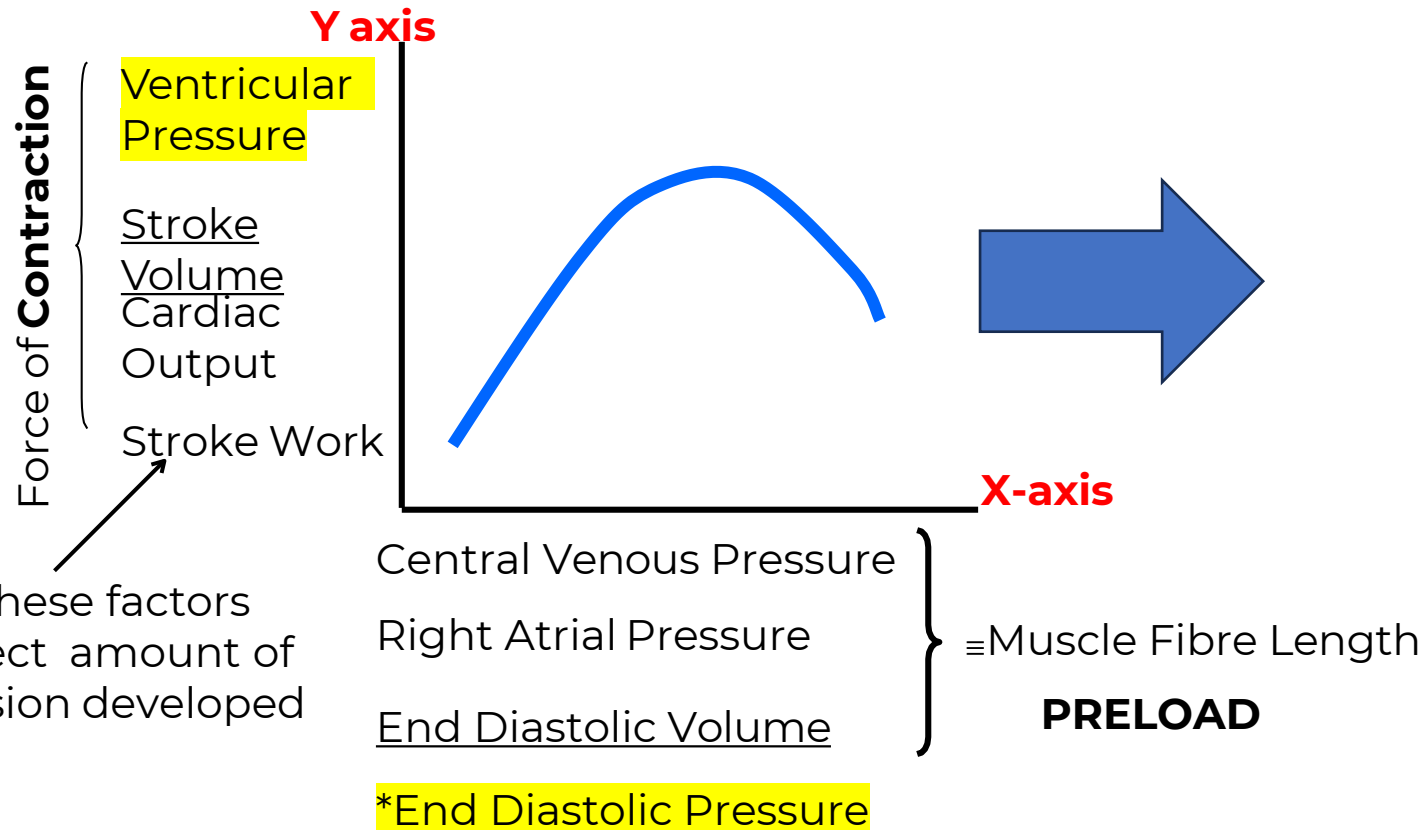
– the relationship between Preload, EDV and SV

- Volume of blood ejected in systole (stroke volume) depends on the volume of blood in the ventricle at the end of the previous diastole (End-Diastolic Volume).
- *EDV is:*
 - *measure of myocardial fibre length*
 - *measure of preload*
 - *determined by venous return*
- *Venous return to right atria is equal to EDV in left ventricle.*
- $\uparrow$ VR increases - $\uparrow$ EDV,
- $\uparrow$ force of contraction - $\uparrow$ stroke volume
- Frank-Starling relationship ensures that the output from the LV and RV are closely matched over time.



$\uparrow$ **PRELOAD ($\uparrow$ EDV)** $\rightarrow$ $\uparrow$ **Contraction** $\rightarrow$ $\uparrow$ **STROKE VOLUME**

Ventricular function 'Curves' – Preload



VENTRICULAR FUNCTION CURVE

Ventricular function curve:

- x-axis is a function of myocardial muscle fibre length
- y-axis is a reflection of contractile energy

Sympathetic activity

– increases ventricular work, cardiac metabolism and cardiac output

Cardiac output

Right Atrial Pressure

CARDIAC PERFORMANCE (FUNCTION) CURVE

Cardiac performance Curve:

- y-axis is Cardiac output
- x-axis is Right atrial pressure

So far:

According to Frank-Starling Law: Stroke volume depends directly on the tension in a stretched myocardial fiber/preload.

Preload is determined by blood volume (EDV) and affected by ventricular compliance

↑ Preload → ↑ SV

How will the other factors will alter SV?

2. Afterload?

3. Contractility?

Ventricular function – Afterload

– the relationship between afterload and SV

Afterload - the stress (tension) the ventricular wall must develop to generate pressures that open the aortic valve and eject blood into the aorta.

To open the aortic valves and eject blood out into the aorta, the LV must develop a tension that generates a pressure slightly greater than the aortic pressure (which is ~80mmHg).

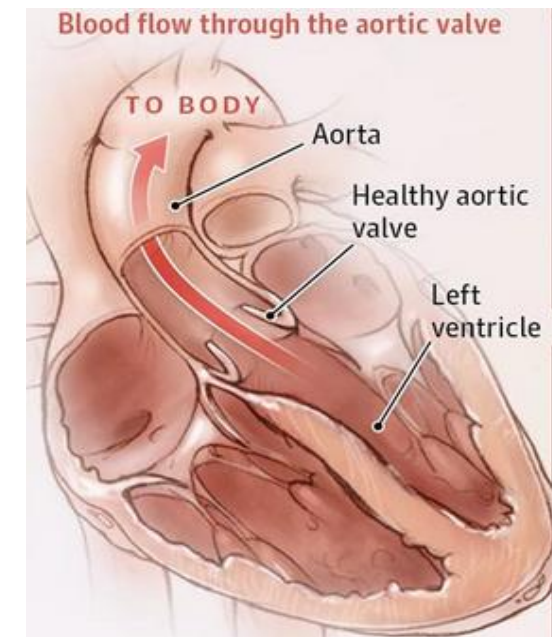
Afterload is often expressed as wall stress. (σ)

Wall stress/tension

consider the heart as a sphere:

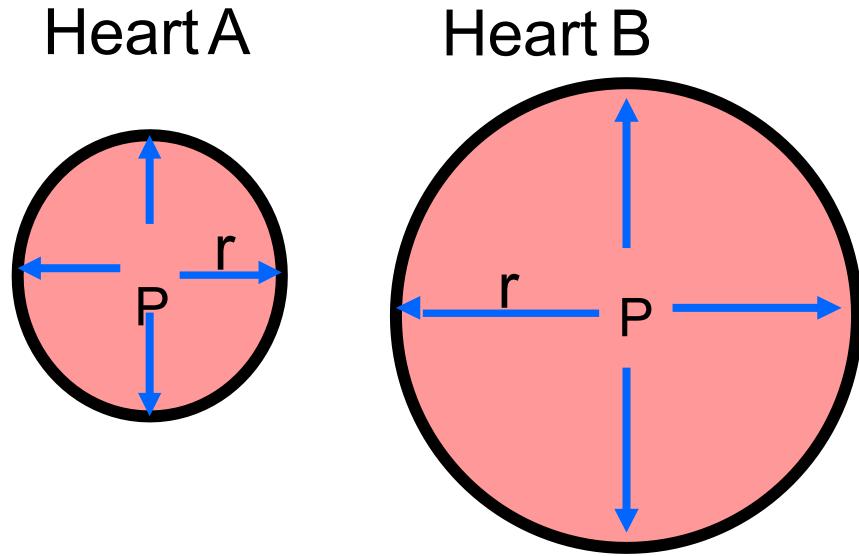
- **tension/wall stress (T) $\propto$ pressure/force (P),**
- **tension/wall stress (T) $\propto$ radius (r)**
- **tension/wall stress (T) $\propto$ 1/thickness (h)**
- **Laplace's Law:** relationship between tension, pressure, radius and wall thickness

$$\text{Tension (wall stress)} = \frac{\text{Pressure x radius}}{2 \times \text{thickness}}$$



Practice this concept:

Afterload/Wall tension and LaPlace's law



$$\text{Tension (wall stress)} = \frac{\text{Pressure} \times \text{radius}}{2 \times \text{thickness}}$$

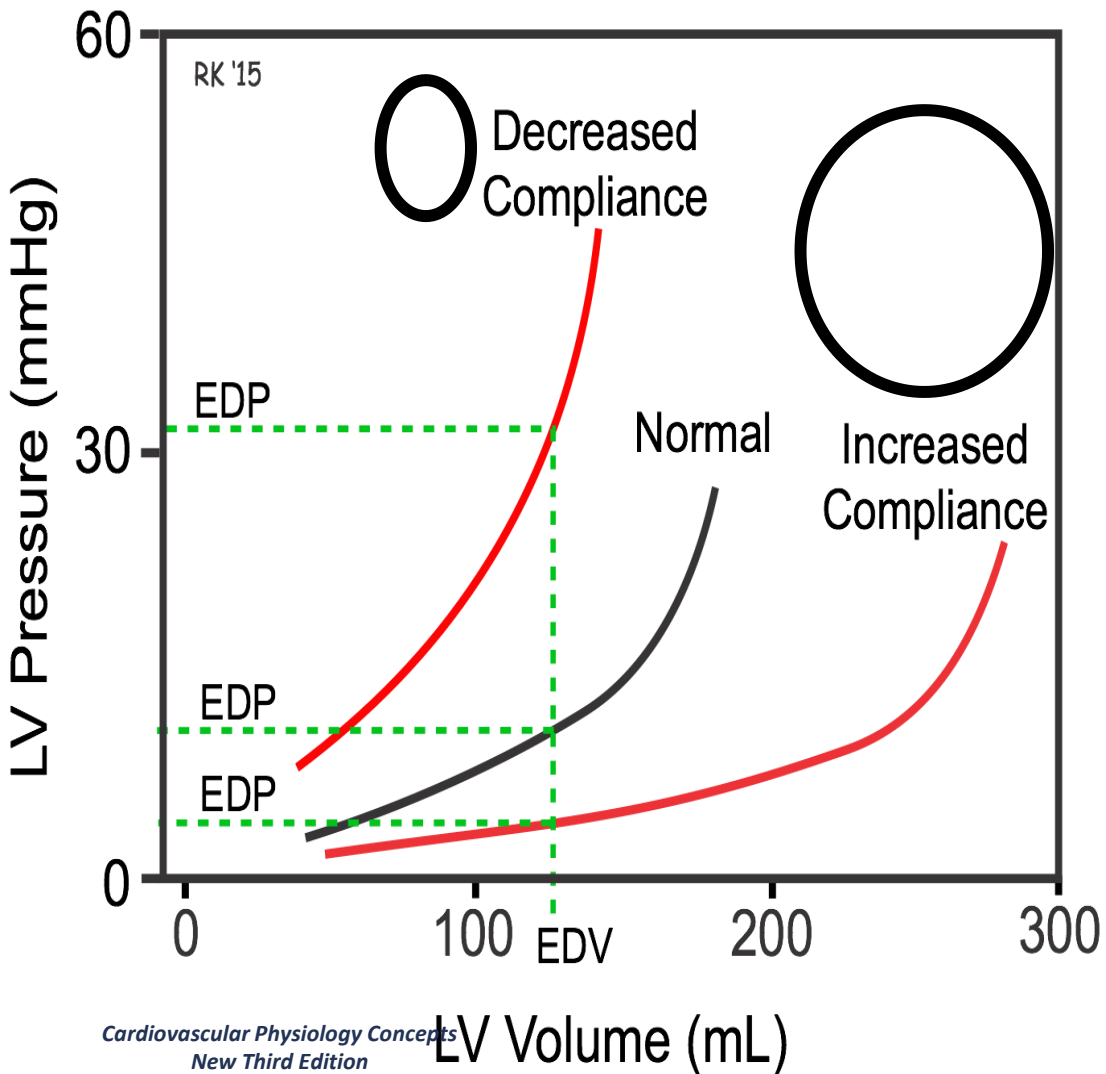
Using LaPlace's law, if given two ventricles with varying chamber radius, *which heart would have the greatest tension?*

To generate the same ventricular pressures in both heart A and B, the ventricle in heart B has to develop greater tension. The “afterload” on heart B is larger.

Clinical significance - a distended heart works harder to eject the same amount of blood into the aorta. In some instances, the ventricles undergo hypertrophy to counteract the increased tension

Ventricular function – Afterload

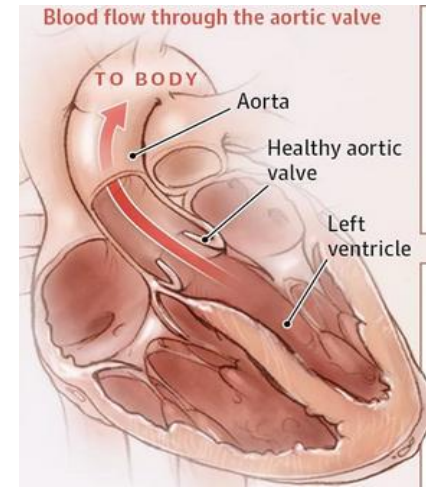
– the relationship between ventricular compliance and SV



- How does ventricular distensibility/compliance affect the end-diastolic pressure?
- How does ventricular distensibility/compliance affect the stroke volume?
- As ventricle fills with blood, the pressure and volume that result from filling are determined by the compliance/distensibility of the ventricle.
- Normal: $\uparrow$ LV EDV – $\uparrow$ LV EDP
- Pathological conditions: LV EDP is related to distensibility/stiffness of myocardium.
- The slope of the relationship is the reciprocal of the compliance, which is sometimes referred to as ventricular "stiffness."
- **$\uparrow$ ventricular stiffness** – $\downarrow$ compliance – $\uparrow$ **LVEDP** (ventricular hypertrophy)
- **$\downarrow$ ventricular stiffness** – $\uparrow$ compliance – $\downarrow$ **LVEDP** (dilated cardiomyopathy)
- A stiffened ventricle has a $\downarrow$ compliance – LV EDP $\uparrow$ **faster**.
- This results in the earlier closure of the mitral valves and $\downarrow$ filling of the heart = $\downarrow$ LV EDV = $\downarrow$ stroke volume.

Ventricular function – Afterload

Factors	Mechanism	Clinical application of concept
Aortic pressure	The pressure that the left ventricle generates during systolic ejection is usually very close to the aortic pressure.	Elevated mean arterial pressure, hypertension - The higher the aortic pressure, the more tension the ventricle must generate. The workload of the heart increased. Compensatory ventricular hypertrophy occurs. Increased wall thickness reduces the amount of tension.
Ventricular dilation	An increase in ventricular chamber radius (ventricular dilation) raises afterload.	Dilated cardiomyopathy
Aortic valve stenosis	Outflow obstruction causes increased left ventricular pressure overload during ejection. It can be much greater than aortic pressure.	Valvular abnormality that causes increased afterload/work of the heart
Systemic vascular resistance/total peripheral resistance	increased vascular resistance and decreased vascular compliance. Determines the diastolic blood pressure – close approximation of afterload	Vasoconstriction increases afterload , increased work of the heart. Vasodilator reduces afterload, reduced work of heart –e.g. 1. Nitroglycerin – relaxation of smooth muscles of blood vessels , 2. Nitric oxide

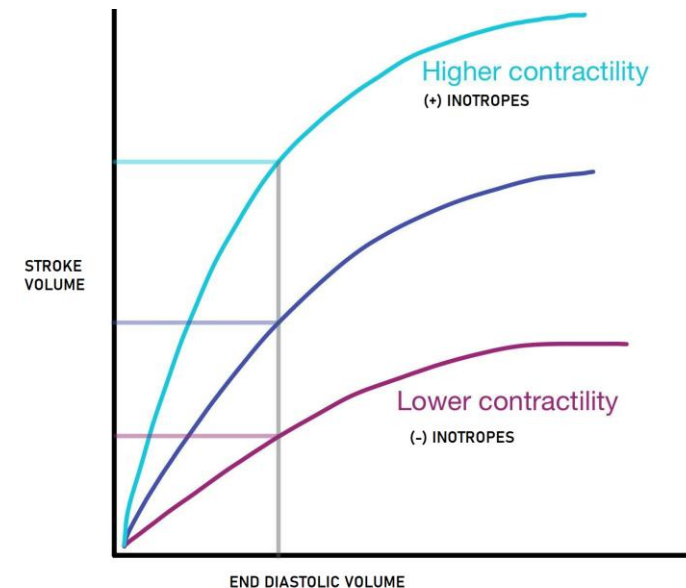


Ventricular function – Contractility

Final
concept

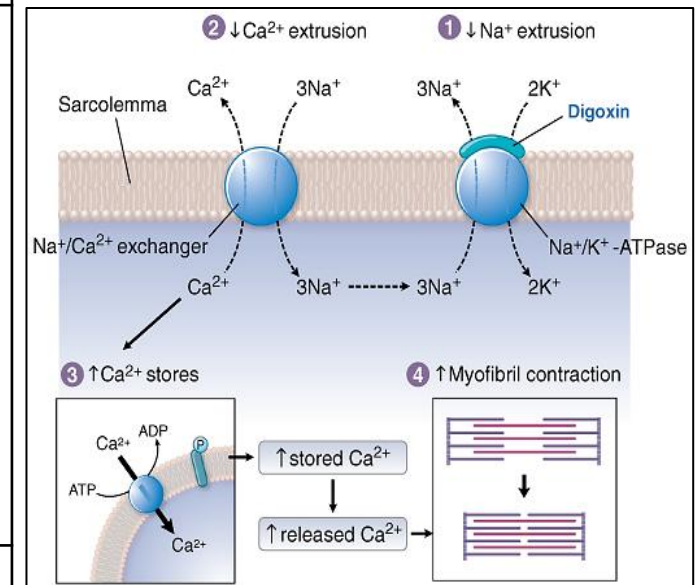
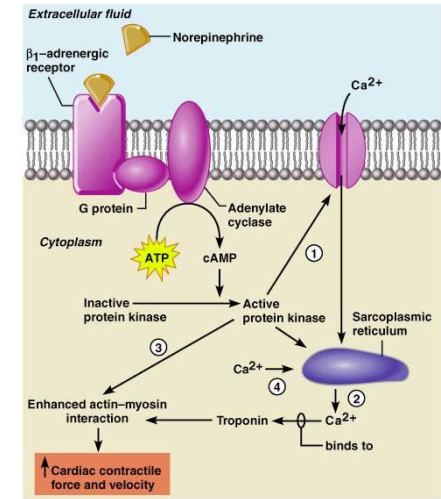
3. CONTRACTILITY (*Inotropism*)

- Acute changes in contractility caused by changes in the intracellular dynamics of calcium; NOT due to changes in fibre length.
- More calcium increases the availability of cross-link sites on the actin; increasing cross-linking and the force of contraction during systole.
- Physiologically, the most important inotrope is sympathetic activity via release of NE (and EPI) acting via β_1 receptors
- Factors or drugs which increase contractility (inotropism) by $\uparrow [Ca^{++}]_i$ via:
 - $\uparrow I_{Ca}$
 - $\uparrow$ Rate of uptake of Ca into SR $\rightarrow \uparrow$ stored Ca in SR
 - Inhibiting Ca – ATPase pump
 - Inhibiting the Na – Ca exchanger
- +ve Inotropes shift the Frank-Starling curve upwards and to the left = $\uparrow$ **SV** at a particular level of **EDV**.
- -ve Inotropes shift curve down and to the right = $\downarrow$ SV



Positive inotropes increase Ca^{2+} availability to improve contractility

POSITIVE Inotrope	Mechanism of action
β receptor agonists: • Isoproterenol/ Dobutamine/ • Norepinephrine • /Epinephrine	cardiac receptors, activate G proteins Adenylate cyclase $\uparrow$ cAMP Protein Kinase A $\uparrow$ i_{Ca} (Phosphorylation) $\uparrow$ Ca transient $\uparrow$ CICR $\uparrow$ Cytosolic Ca
Cardiac glycosides Digoxin	• Digoxin inhibits Na-K ATPase • $\uparrow$ Intracellular Na • Decreased extrusion of Ca^{2+} by the Na^+ – Ca^{2+} exchanger • More cytosolic Ca^{2+} • More Ca^{2+} pumped back into SR • More Ca^{2+} available in the SR for cardiac contraction • $\uparrow$ contractility of the heart
Caffeine	• inhibits PDE – thereby $\uparrow$ cAMP, exerts a positive inotropic (and chronotropic effect)



Negative inotropes reduce Ca^{2+} availability to reduce contractility

Negative Inotrope	Example	Mechanism of action
Drugs	Calcium channel blockers: (Nondihydropyridines) Verapamil	L-type calcium channel blocker decreases the strength of myocardial contraction
	β receptor Blockers: Propranolol	non-selective beta-adrenergic antagonist
Cardiac Pathology	Myocardial infarction/ischemia	$\uparrow [\text{H}^+]$ H^+ compete with Ca^{2+} for binding to troponin (OR $\downarrow$ Affinity of TN-C for Ca^{2+})
	Cardiac Failure	$\downarrow$ Expression of Ca-ATPase pump $\downarrow$ Ca stored in SR $\downarrow$ Ca available for contraction $\downarrow$ Contractility

- Calcium dynamics does not explain chronic losses in contractility, which in most cases are due to overall myocyte dysfunction.

How is cardiac contraction regulated?

Target various aspects of calcium release and re-uptake ...

Cardiac muscle contraction (Inotropy)	
Ca entry into myocytes	The major factor that regulates Ca entry into cells is norepinephrine (or epinephrine) acting upon β_1 receptors. The resulting increase in cAMP activates PKA which phosphorylates L-type Ca channels thus increasing their permeability to Ca which increases Ca influx.
Ca release from SR	Activation of β_1 receptors via increased PKA will also result in phosphorylation of sites on the SR which leads to increased Ca release.
Ca Binding to TN-C	Any factor that increases intracellular Ca will result in a greater amount of Ca being bound to TN-C – resulting in a greater force developed. Note that in some situations such as acidosis, resulting from myocardial hypoxia, the affinity of Ca for TN-C is actually reduced, and this then has a negative inotropic effect. Changes in myocardial fibre length also increases the sensitivity of TN-C to Ca, although the actual mechanism is unknown. (Myocardial muscle fibre length is akin to preload; therefore, increasing preload increases contraction because of the increased binding of Ca to TN-C).
Ca Uptake by SR	Increasing Ca uptake into the SR by SERCA indirectly increases the amount of Ca released by the SR – the more Ca that is sequestered the more is available for subsequent release. PK-A phosphorylates phospholamban thus removing its inhibitory effect on SERCA.
Regulation of Ca Efflux	Ca is extruded from the cytosol to the extracellular fluid by the Na-Ca exchanger (which is dependent on the Na pump) and is resequenced back into the SR by SERCA. Factors which affect the Na-Ca exchanger will then affect cytosolic Ca levels. The cardiac glycoside, digoxin inhibits the Na pump which impairs the functioning of the Na-Ca exchanger resulting in less Ca being expelled and more Ca staying in the cytosol. Contraction is then enhanced.

Therefore, **sympathetic activity via activation of β_1 receptors** (inotropic agents (β_1 agonists) **can increase inotropy** by several different mechanisms which are listed above.